



Faculty of Software Engineering and Computer Systems

Programming

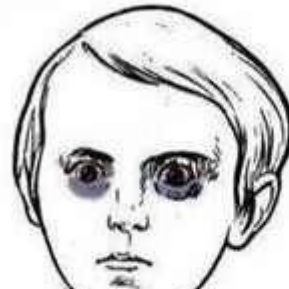
Lecture #0
Introduction and more.

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Introduction. Course goals

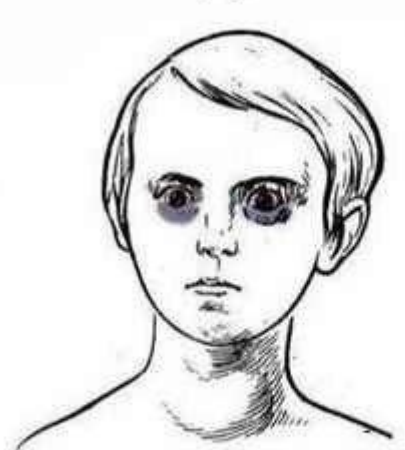
- Learn Java from basics to advanced technology
- In this course, you will acquire programming skills at a sufficient level to work in IT
- After this course, you will be able to stay awake for two nights in a row and feel ~~great~~



Introduction. Course objectives

After this course, you should be able to do the following:

- Create simple and complex Java-applications
- Learn to apply programming paradigms such as inheritance, encapsulation, and polymorphism. This skill is useful in programming in any language
- Learn to compile, build and run Java applications of any complexity
- ...
- stay awake for two nights in a row and feel great



Introduction. Prerequisites

To successfully complete this course, you must know:

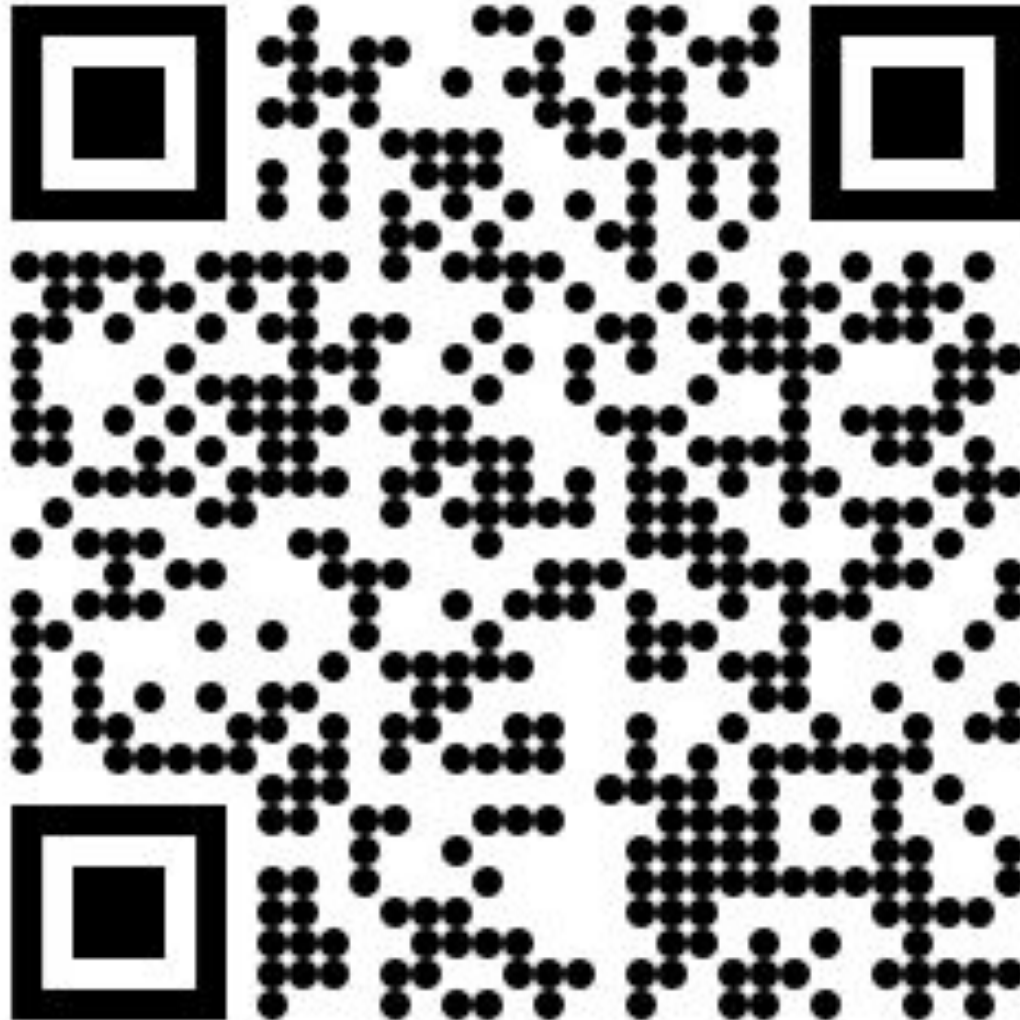
- basics of programming and informatics
- about two hours of time each day for this course
- how to stay awake **one** night and feel great



Introduction. Materials and news



Introduction. Lectures (partly)



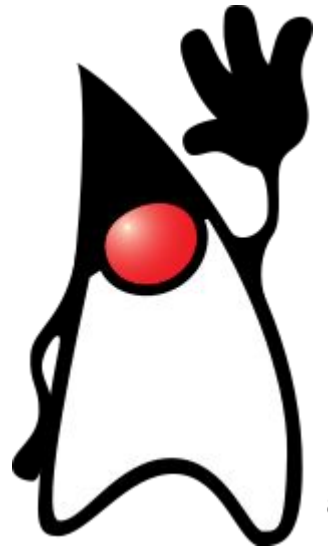
Introduction. Course parts

- How much labs
- How much tests
- Lectures
- Exams

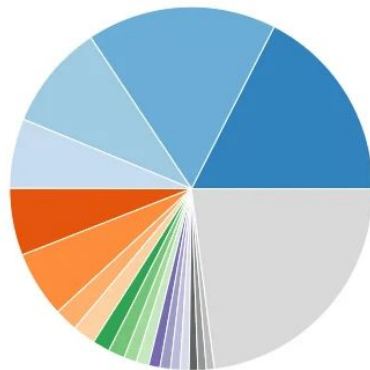
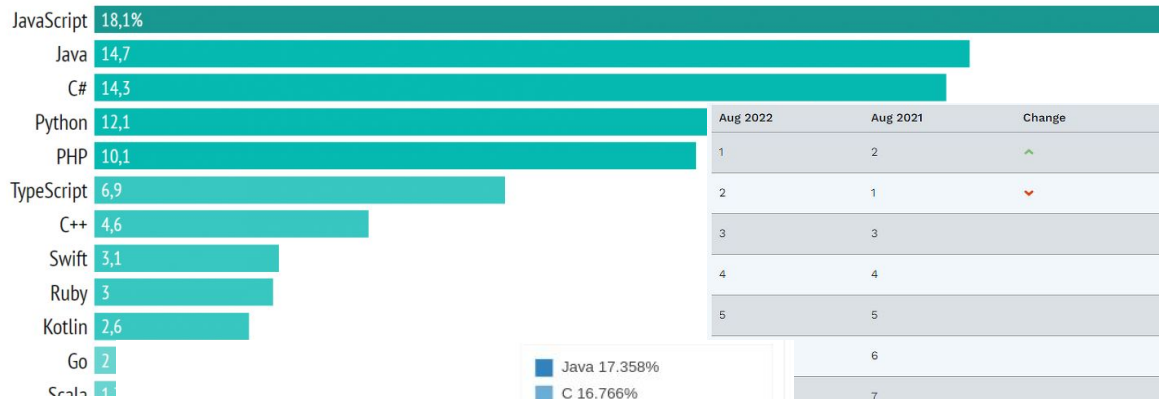
Introduction. Tools

- Text editor (SublimeText, Notepad++)
- JDK
- IDE (intelliJ IDEA, Eclipse, NetBeans)
- CMD/Terminal (Putty, WinSCP, shell, helios)

Why Java?



Introduction. Why Java?



	Aug 2022	Aug 2021	Change	Programming Language	Ratings	Change
1	2	1	▲	Python	15.42%	+3.56%
2	1	2	▼	C	14.59%	+2.03%
3	3	3		Java	12.40%	+1.96%
4	4	4		C++	10.17%	+2.81%
5	5	5		C#	5.59%	+0.45%
6				Visual Basic	4.99%	+0.33%
7				JavaScript	2.33%	-0.61%
9			▲	Assembly language	2.17%	+0.14%
10			▲	SQL	1.70%	+0.23%
8			▼	PHP	1.39%	-0.80%

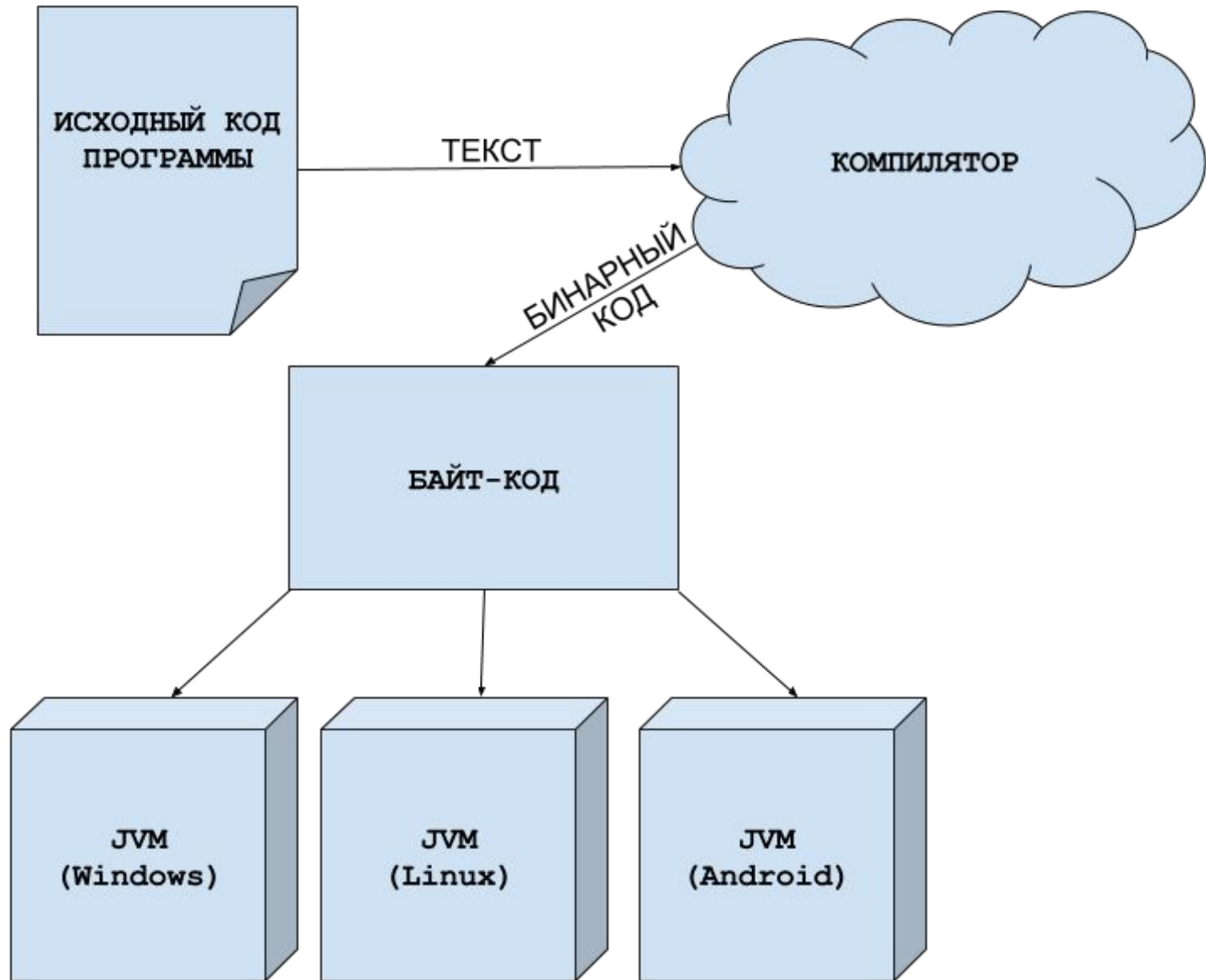
Rank	Language	Type	Score
1	Python	🌐 📱 📺	100.0
2	Java	🌐 📱 📺	95.3
3	C	📱 📺 🌐	94.6
4	C++	📱 📺 🌐	87.0
5	JavaScript	🌐	79.5
6	R	📺	78.6
7	Arduino	🌐	73.2
8	Go	🌐 📱 📺	73.1
9	Swift	📱 📺	70.5
10	Matlab	📺	68.4

Introduction. Why Java?

- high threshold for entry into IT (low competition)
- Java is a very large and complex product. Knowing Java is easy to master other areas
- cross-platform



Introduction. Cross-platform



Introduction. Cross-platform

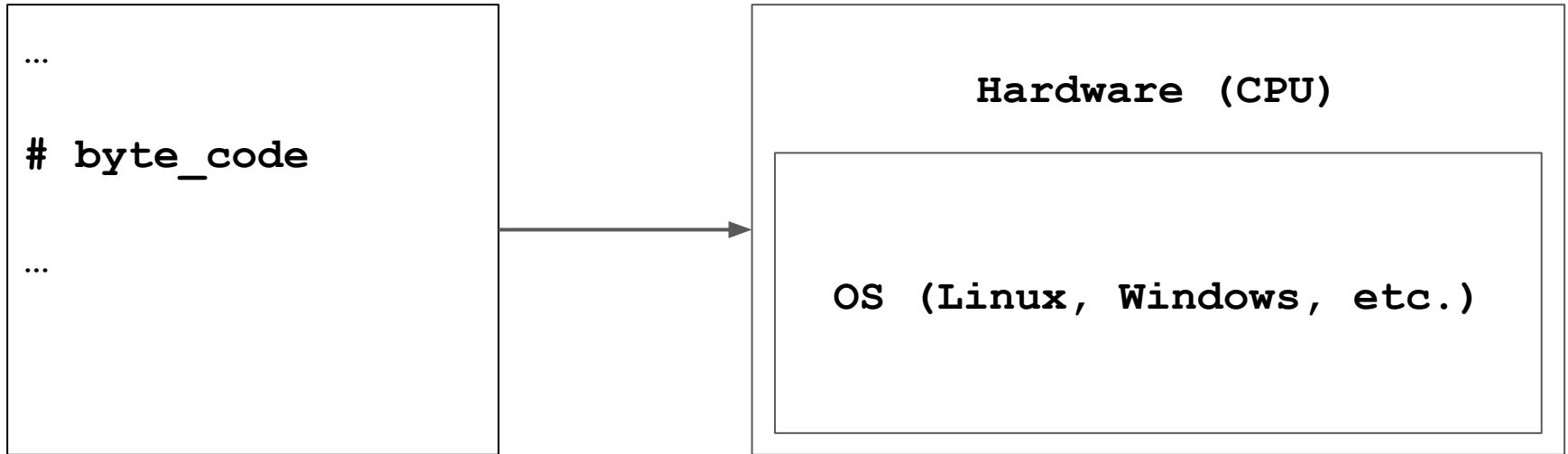
```
public class Object
    public Object() {}

    ...
}
```



```
Compiled from "Object.java"
public class java.lang.Object
{
    public java.lang.Object();
    Code:
        0: return
    public boolean
    equals(java.lang.Object);
    Code:
        0: aload_0
        1: aload_1
        2: if_acmpne      9
        5: iconst_1
        6: goto           10
        9: iconst_0
       10: ireturn
    ...
}
```

Introduction. Cross-platform



НИЧОСИ



Java

Java. Comments

Однострочные комментарии

// комментарии

Многострочные комментарии

/
комментарии
/

Специальные комментарии (javadoc)

*/**
*
* @author James Gosling
/

Java. First application

Main.java

```
1. public class Main {  
2.  
3.     public static void main(String[] args) {  
4.  
5.         System.out.println("Внимание!");  
6.  
7.     }  
8.  
9. }
```



Java. Simple math-application

```
1.  public class SimpleProgram {  
2.  
3.      public static void main(String[] args) {  
4.  
5.          var x = 5.0D;  
6.  
7.          var result = Math.pow(x, 2.5);  
8.  
9.          System.out.println(result);  
10.  
11.      }  
12.  
13. }
```

Java. Program is written. What's next?

JDK

- compiler
- debugger
- byte-code tools
- JVM
- standard library

Java. Program is written. What's next?

1. Compilation

```
javac    Main.java
```

```
javac    -d    target    Main.java
```

```
javac    --help
```

2. Packaging to jar

```
jar      -cfm    app.jar    MANIFEST.mf    *.class
```

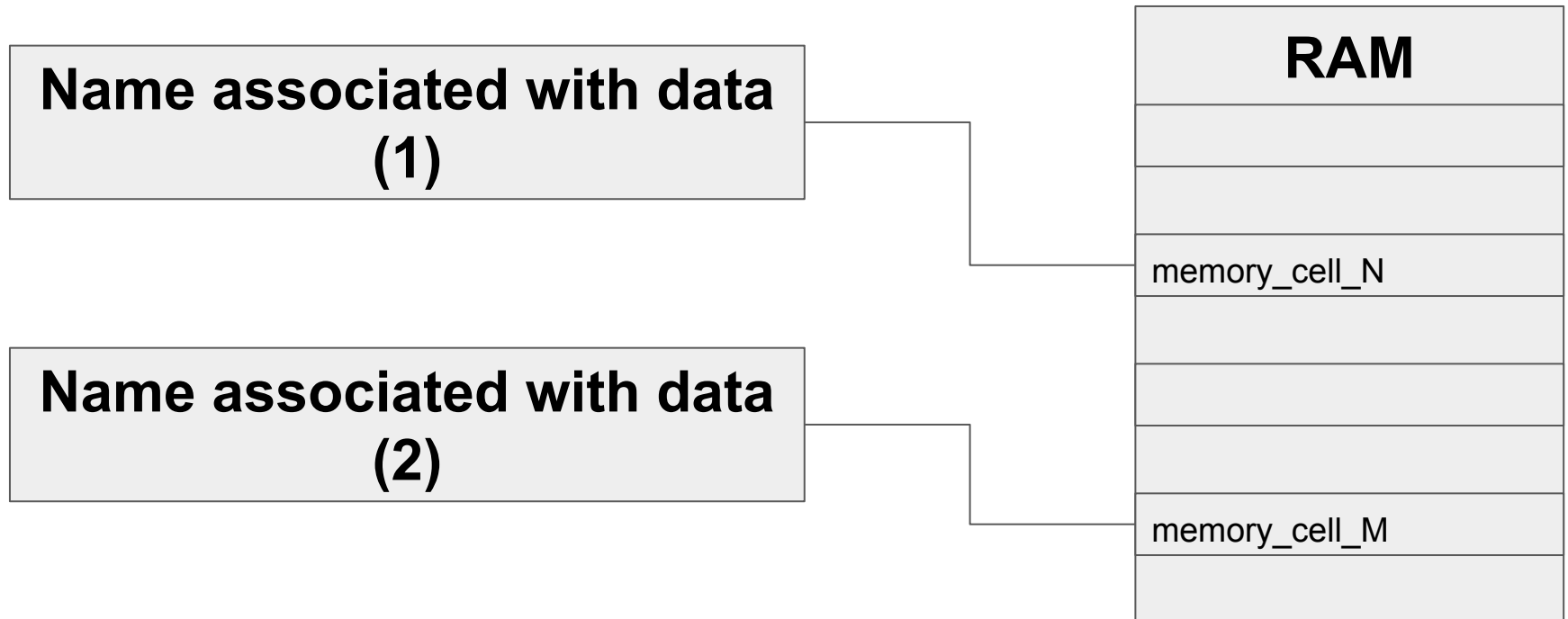
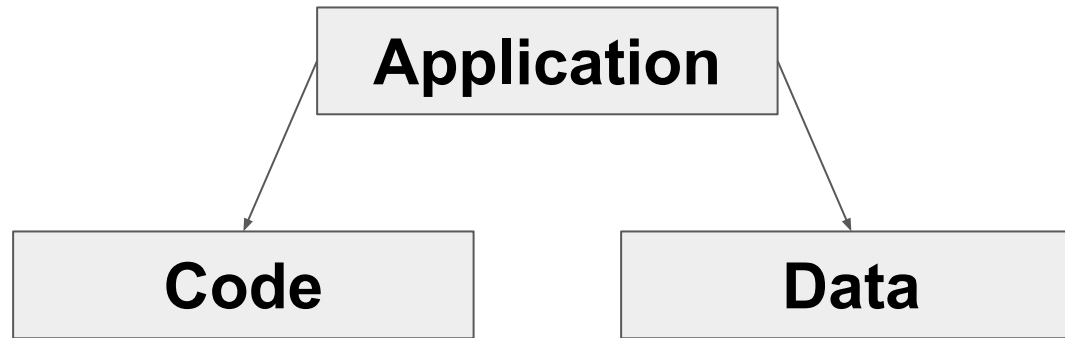
3. Run

```
java     -jar    app.jar
```

```
java     Main
```

4. Debugging

Java. Variables and constants

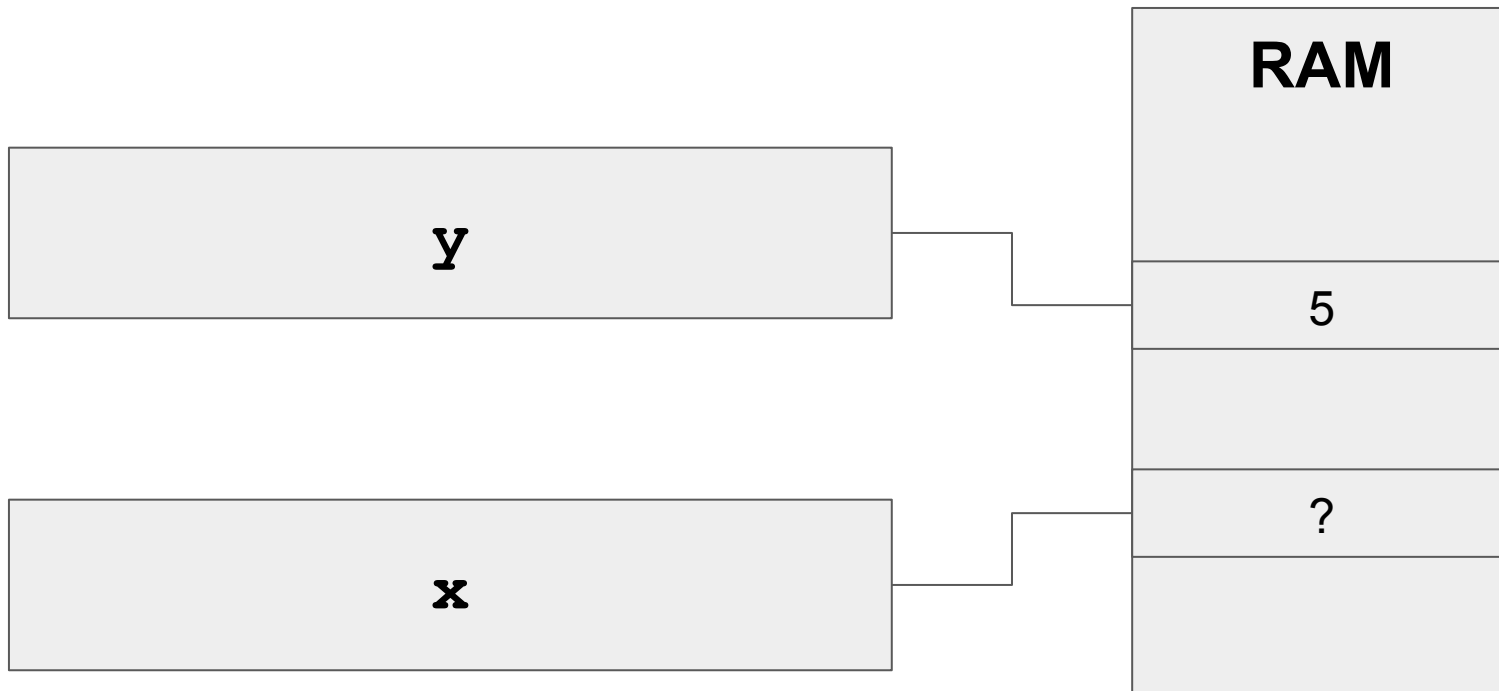


Java. Variables and constants

```
int x;
```

```
int y = 5;
```

```
data_type variable_name [ = default_value ] ;
```



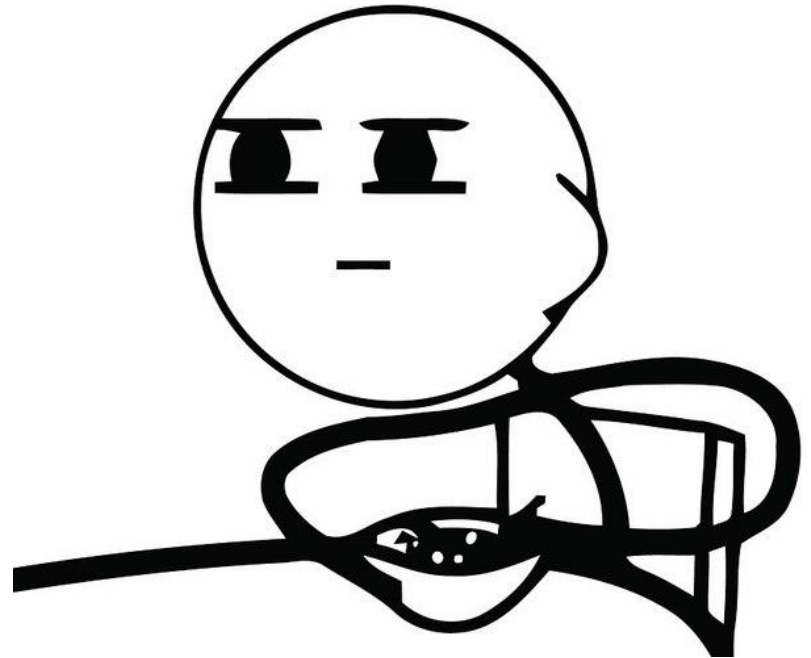
Java. Variables and constants

```
final double E = 5.0;
```

```
final double R;  
// ...  
R = 5.0;
```

what is it for?

```
int total = 8 * 5 * 4 * 12;
```



Java. Variables and constants

what is it for?

```
int total = 8 * 5 * 4 * 12; // BAD
```

```
final int HOURS = 8, DAYS = 5, WEEKS = 4, MONTHS = 12;  
// ...  
int total = HOURS * DAYS * WEEKS * MONTHS; // GOOD
```



Java. Data types

Целочисленные

Вещественные

Логические

Объекты

byte

double

boolean

short

float

char

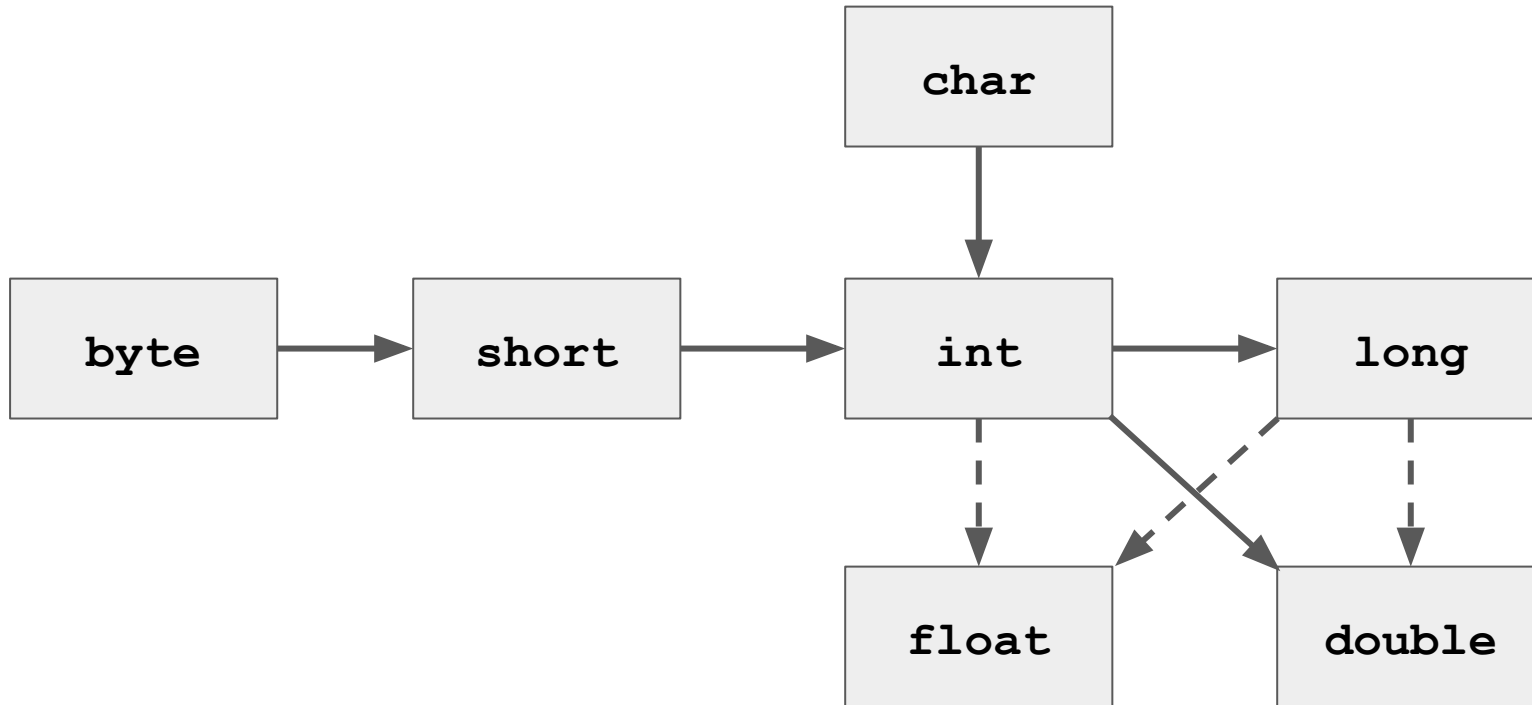
int

long

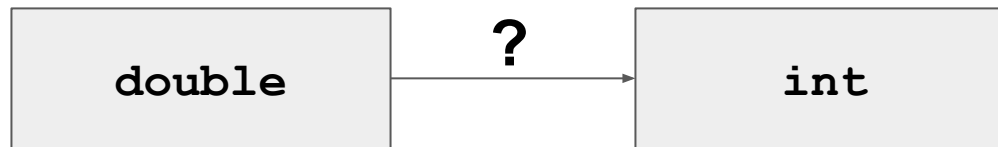
Java. Data types

Data type	Size (bit)	Values
byte	8	○T -128 до 127
short	16	○T -32768 до 32767
char	16	○T 0 до 65535
int	32	○T -2147483648 до 2147483647
long	64	○T -9223372036854775808 до 9223372036854775807
float	32	○T -1.4e-45f до 3.4e+38f
double	64	○T -4.9e-324 до 1.7e+308
boolean	1 or 32	true or false

Java. Convert data types



Java. Casting data types



```
double e = 2.7;
```

```
int notE = (int) e;    // notE = 2
```

```
int roundE = (int) Math.round(e);    // roundE = 3
```

Java. Operators

Unary

-
--
++
~
!
[]

Binary

- +

/ *

| ||

& &&

== !=

```
int x = -y;
```

```
// унарный "минус"
```

```
int r = x - y;
```

```
// бинарный "минус"
```

Java. Operators

№	Operator
1	[] . ()
2	! ~ ++ -- + - (cast) new
3	* / %
4	+ - (binary)
5	>> << >>>
6	< <= > >= instanceof
7	== !=
8	&
9	^
10	
11	&&
12	
13	?:
14	= += -= *= /= %= = ^= <<= >>= >>>=

Java. Math functions

Функция	Конструкция
Абсолютное значение x	<code>Math.abs (x);</code>
Косинус x	<code>Math.cos (x);</code>
Синус x	<code>Math.sin (x);</code>
Экспонента	<code>Math.exp (x);</code>
Квадратный корень x	<code>Math.sqrt (x);</code>
Корень суммы квадратов (гипотенуза)	<code>Math.hypot (x, y);</code>
Натуральный логарифм	<code>Math.log (x);</code>
Округление до ближайшего	<code>Math.round (x);</code>
Возведение 'x' в степень 'y'	<code>Math.pow(x, y);</code>

<https://docs.oracle.com/javase/8/docs/api/java/lang/Math.html>

Java. Math constants

- `Math.PI` `// 3.14159265358979323846`
- `Math.E` `// 2.7182818284590452354`

```
double res1 = 3 * Math.PI / 2;
```

```
double res2 = Math.log(5) * Math.E;
```

Java. Just in time test

```
1. public class PracticValiables {
2.
3.     public static void main(String[] args) {
4.
5.         byte x = 127;
6.
7.         x++;
8.
9.         System.out.println(x) ;
10.
11.     }
12. }
```

128

1

-1

127

Java. Lines

```
String name = "Alis";
```

```
String lastName = new String("Parker");
```

```
String a = "Java\u2122";           // Java™  
String b = "";                   // пустая строка
```

<https://docs.oracle.com/javase/8/docs/api/java/lang/String.html>

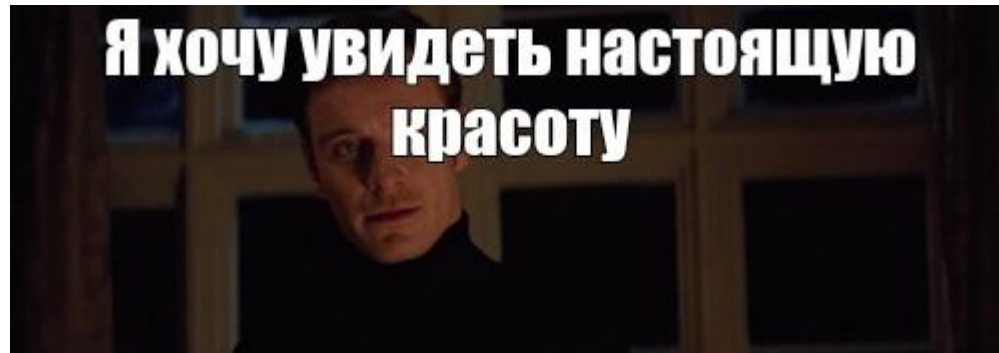
Java. Formatting output

```
double x = 5.683;
```

```
double y = Math.pow(x, 2.5);
```

```
System.out.printf("%.2f ^ 2.5 = %.3f", x, y);
```

Output: 5,68 ^ 2.5 = 76,890



Java. Formatting output

```
printf(String format, Object... args)
```

```
printf("%.4f", Object... args)
```

What is "format"?

```
%[argument_index$][flags][width][.precision]conversion
```

Эльфийско-русский словарь

%[argument_index\$][flags][width][.precision]conversion

argument_index\$ — целое десятичное число, указывающее позицию аргумента в списке аргументов

flags — специальные символы для форматирования. Например, флаг "+" означает, что числовое значение должно включать знак +, флаг "-" означает выравнивание результата по левому краю, флаг «,» устанавливает разделитель тысяч у целых чисел.

[width] — положительное целое десятичное число, которое определяет минимальное количество символов, которые будут выведены

[.precision] — неотрицательное целое десятичное число с точкой перед ним. Обычно используется для ограничения количества символов.

conversion — это символ, указывающий, как аргумент должен быть отформатирован. Например **d** для целых чисел, **s** для строк, **f** для чисел с плавающей точкой.

* copy-paste from Java documentation

Java. Summary about you next steps

- You will do the lab work at home
 - Write code
 - Copy it to server (helios.se.ifmo.ru)
 - Compile and run it
- Your teacher may give you additional practice tasks
- Examination of lab work
 - Be ready to hear the question, make up from unfamiliar words
 - Try to find answer and repeat discuss with teacher

